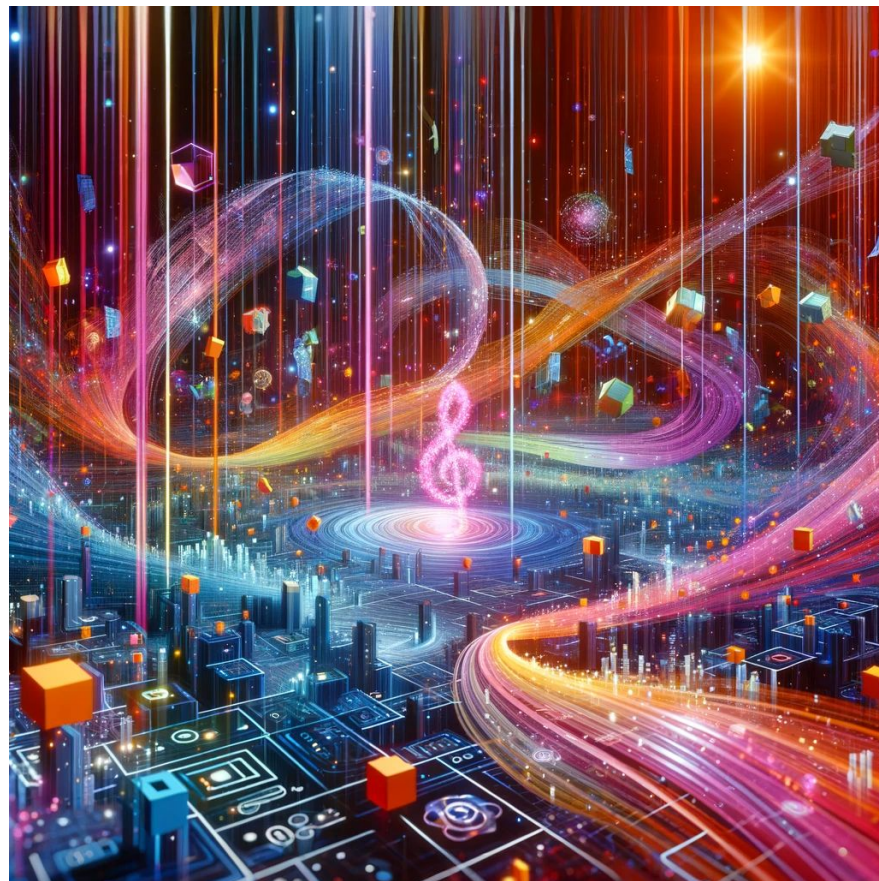
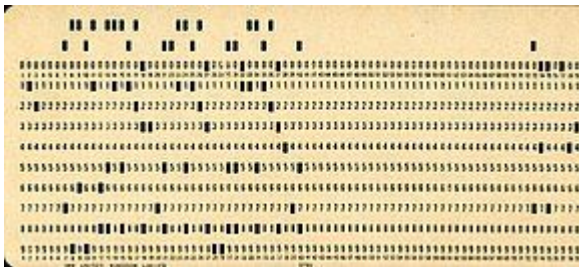


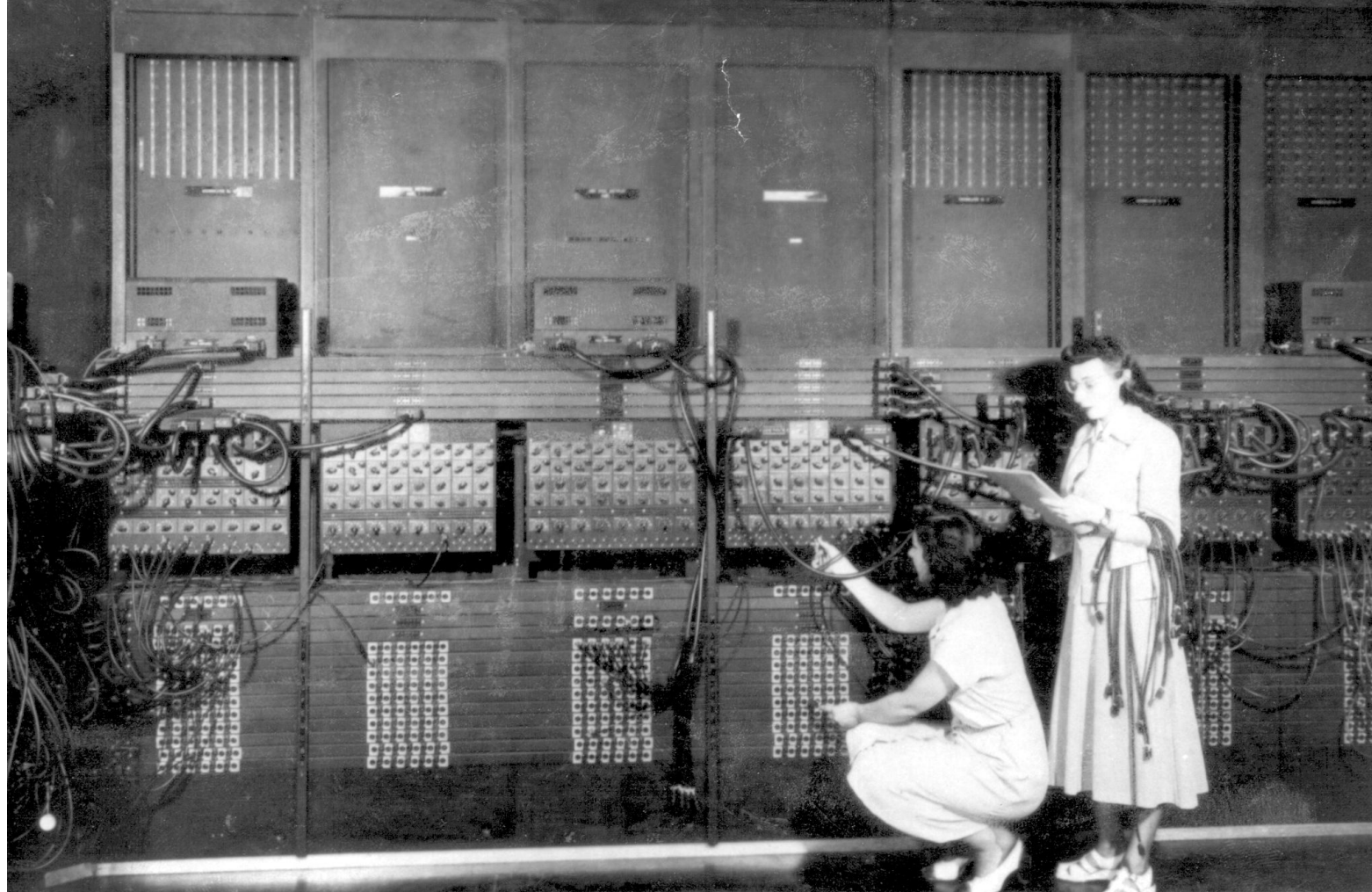
SPRING 2026
DSE I2450 - 3GG
BIG Data &
Scalable Computation

Week 1:
Distributed File Systems
(GFS + HDFS)



Dall-E 3: "inside a big data system, use your imagination"





ENIAC programmers, [source](#)



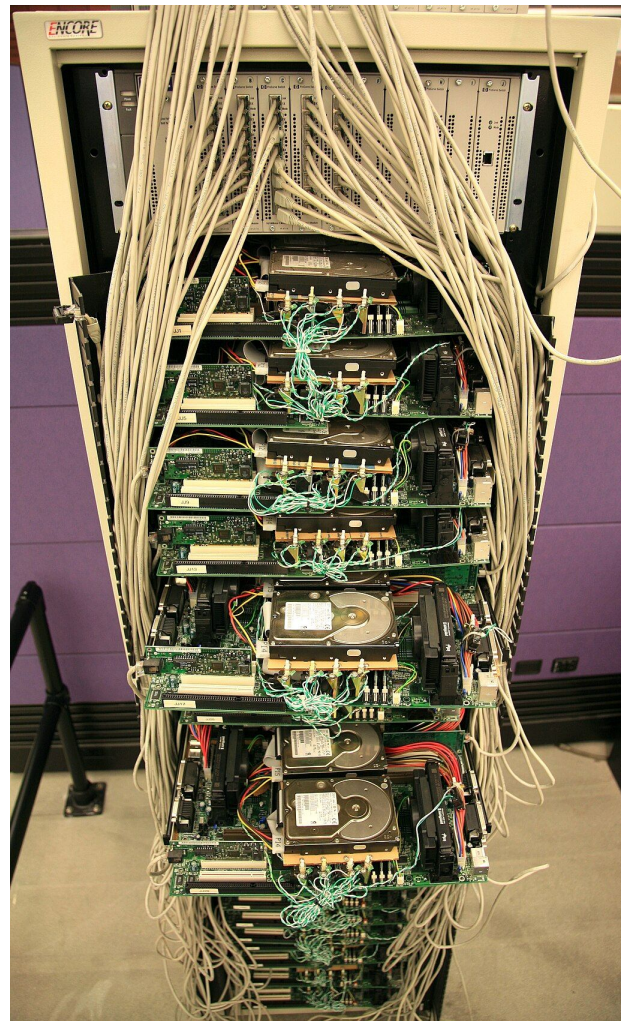
Dall-E 3: library of Alexandria, “inside with lots of information”

Google's first
"production server"!
c. 1999

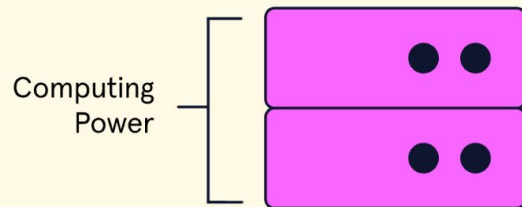
so ... scalability

source: wikipedia

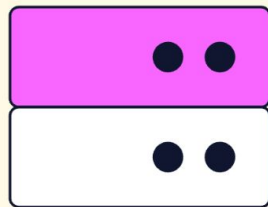
https://en.wikipedia.org/wiki/Google#/media/File:Google%E2%80%99s_First_Production_Server.jpg



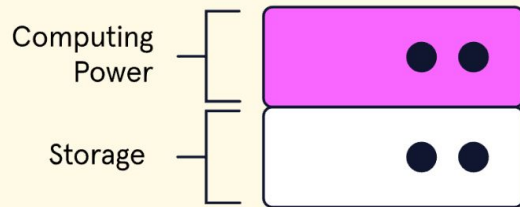
Cluster Manager



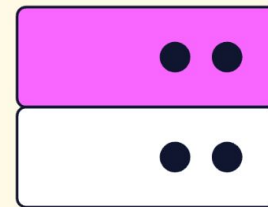
Commands



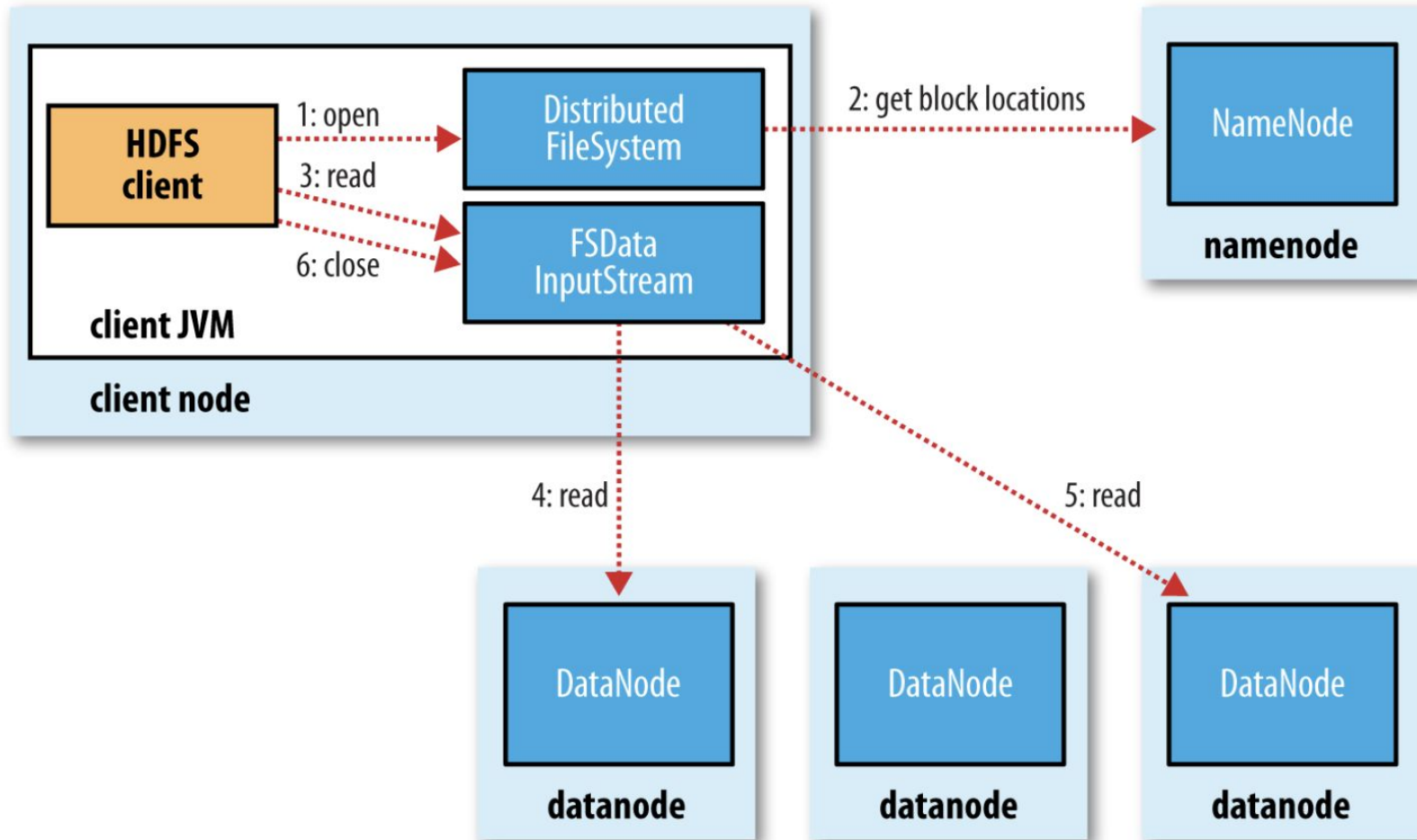
Worker Node



Worker Node



Worker Node



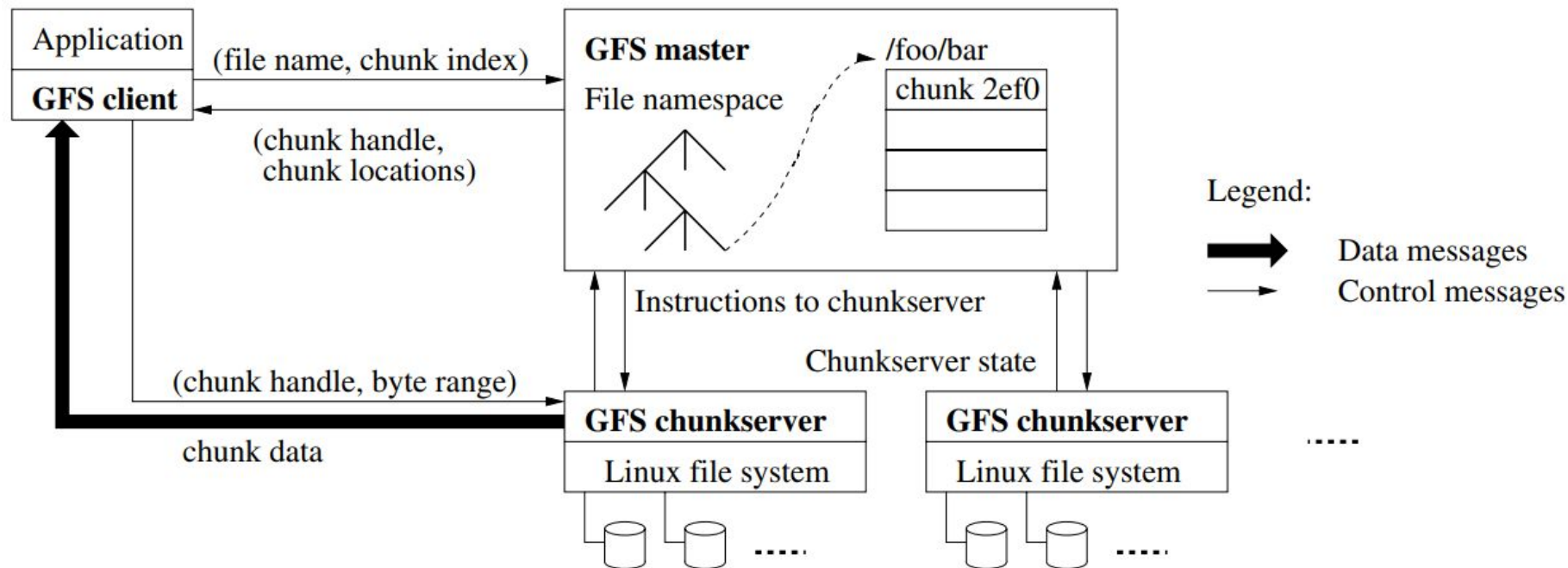


Figure 1: GFS Architecture

For example, imagine a node $n1$ on rack $r1$ in data center $d1$. This can be represented as $/d1/r1/n1$. Using this notation, here are the distances for the four scenarios:

- $distance(/d1/r1/n1, /d1/r1/n1) = 0$ (processes on the same node)
- $distance(/d1/r1/n1, /d1/r1/n2) = 2$ (different nodes on the same rack)
- $distance(/d1/r1/n1, /d1/r2/n3) = 4$ (nodes on different racks in the same data center)
- $distance(/d1/r1/n1, /d2/r3/n4) = 6$ (nodes in different data centers)

This is illustrated schematically in [Figure 3-3](#). (Mathematically inclined readers will notice that this is an example of a distance metric.)

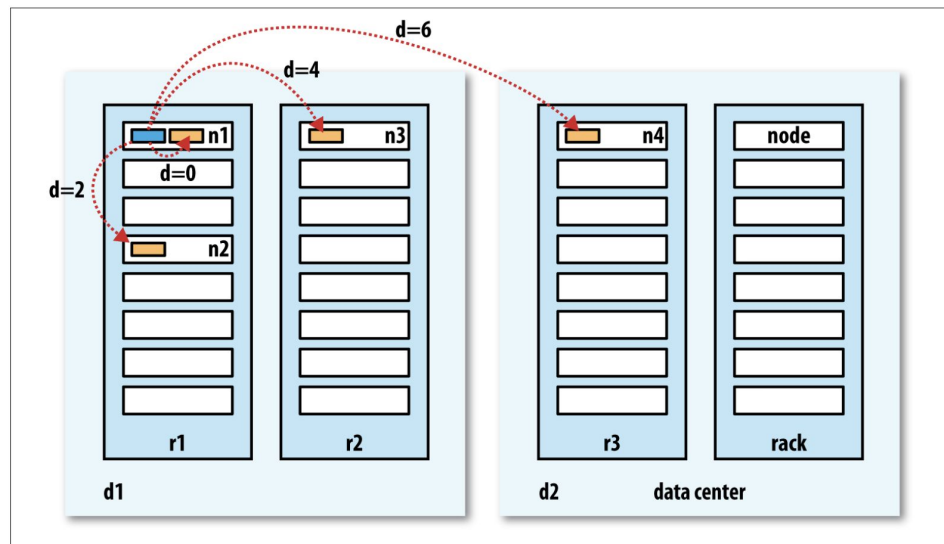


Figure 3-3. Network distance in Hadoop



The systematic looting of language can be recognized by the tendency of its users to forgo its nuanced, complex, midwifery properties for menace and subjugation. Oppressive language does more than represent violence; it is violence; does more than represent the limits of knowledge; it limits knowledge.

- Toni Morrison

ON "disk" = hard drive
maintains memory when power = OFF

SLOWER THAN RAM!

random access memory
RAM remembers when ON

Hard drive like a record player ...
mechanical moving parts

RAM = like a grid of cells



0236

1-1

46V16M8

TG

-6T B

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-6T B

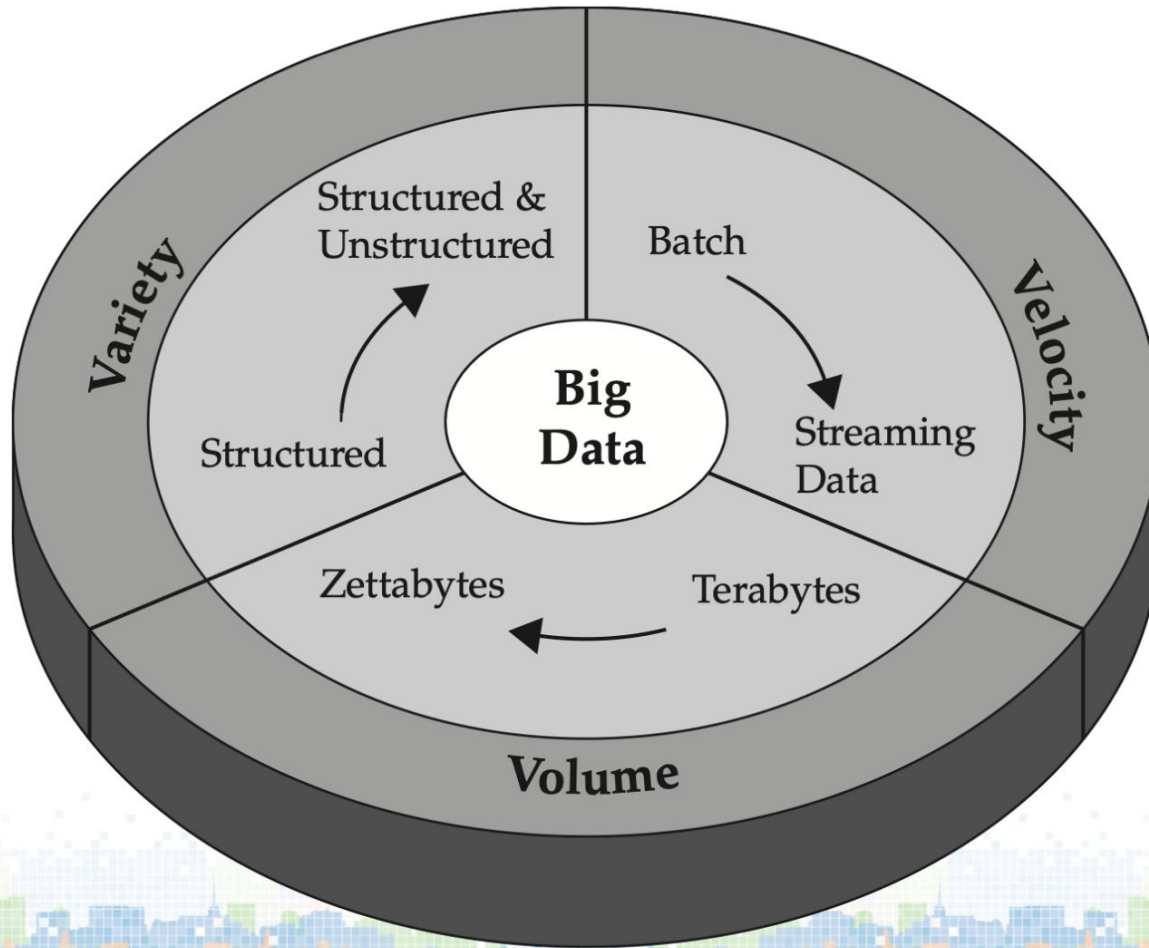
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TG

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[Source: IBM, 2012]
[Source: MapR 2014]

